

Q1

```
public class Book {

    enum Status {
        ALREADY_READ,
        READING,
        TO_BE_READ
    }

    private int id;
    private String name;
    private String author;
    private int rank;
    private Status status = Status.TO_BE_READ;

    public Book() {

    }

    public Book(String name, String author, int rank) {
        this.rank = rank;
        this.name = name;
        this.author = author;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getName() {
        return this.name;
    }

    public void setAuthor(String author) {
        this.author = author;
    }

    public String getAuthor() {
        return this.author;
    }

    public void setId(int id) {
        this.id = id;
    }
}
```

```

    public int getId() {
        return this.id;
    }

    public void setRank(int rank) {
        this.rank = rank;
    }

    public int getRank() {
        return this.rank;
    }

    public void setStatus(String stts) {
        switch (stts) {
            case "ALREADY_READ":
                this.status = Status.ALREADY_READ;
                break;
            case "READING":
                this.status = Status.READING;
                break;
            case "TO_BE_READ":
                this.status = Status.TO_BE_READ;
                break;
        }
    }

    public String getStatus() {
        return this.status.name();
    }
}

```

```

import java.util.Scanner;

```

```

public class BookList {

    private String bookListName;
    public Book[] bookList = new Book[15];
    private int bookCount = 0;

    public BookList(String listname) {

        this.bookListName = listname;
    }

    Scanner sc = new Scanner(System.in);

```

```

public String getListName() {
    return bookListName;
}

```

//Adding a book

```

public int AddBook(String name, String author, int rank) {

    bookList[bookCount] = new Book(name, author, rank);
    bookList[bookCount].setId(bookCount + 1);
    bookCount++;
    return bookCount;
}

```

//Deleting book

```

public int DeleteBook(int id) {

    for (int i = id; i <= bookCount - 1; i++) {
        if (i == 15) {
            bookList[14] = null;
            break;
        }
        bookList[i - 1] = bookList[i];
        bookList[i - 1].setId(i);
    }

    bookCount--;
    return bookCount;
}

```

//Display book list

```

public void DisplayList() {
    System.out.println("ID" + "\tName" + "\t\tAuthor(s)" + "\tRank" + "\tStatus");
    for (int l = 0; l <= bookCount - 1; l++) {

        System.out.println(bookList[l].getId() + "\t" + bookList[l].getName() + "\t\t" +
bookList[l].getAuthor()
            + "\t" + bookList[l].getRank() + "\t"
            + bookList[l].getStatus());
    }
}

```

// Displaying a book by name

```

public void DisplayBook(String name) {
    for (Book book : bookList) {
        if (book.getName() == name) {
            if (book != null) {
                System.out.println("ID : " + "\t" + book.getId());
            }
        }
    }
}

```

```

        System.out.println("Name : " + "\t" + book.getName());
        System.out.println("Author : " + "\t" + book.getAuthor());
        System.out.println("Rank : " + "\t" + book.getRank());
        System.out.println("Status : " + "\t" + book.getStatus());
    }
}
}
}
// search a book
public Book searchBook(String name) {
    for (Book book : bookList) {
        if (book.getName() == name) {
            return book;
        }
    }
    return null;
}

//To modify
public void ModifyBook(String name) {
    for (Book book : bookList) {
        if (book.getName() == name) {
            System.out.println("Select Option");
            System.out.println("1. Modify Name");
            System.out.println("2. Modify Author(s)");
            System.out.println("3. Modify Rank");
            System.out.println("4. Modify Status");
            int choice = sc.nextInt();
            switch (choice) {
                case 1:
                    System.out.println("Enter new name");
                    String new_name = sc.nextLine();
                    book.setName(new_name);
                    break;

                case 2:
                    System.out.println("Enter Authors");
                    String author_1 = sc.nextLine();
                    book.setAuthor(author_1);
                    break;

                case 3:
                    System.out.println("Enter Authors");
                    int rnk = sc.nextInt();
                    book.setRank(rnk);
                    break;

                case 4:
                    System.out.print("Enter Status");
                    System.out.println("ALREADY_READ , READING, TO_BE_READ ");

```

```

        String stts = sc.nextLine();
        book.setStatus(stts);
        break;

    default:
        System.out.println("Invalid input");
        break;
    }
}
}
}

```

//Get current reading book

```

public void getCurrentBook() {
    for (Book book : bookList) {
        if (book.getStatus() == "READING") {
            System.out.println("ID : " + "\t" + book.getId());
            System.out.println("Name : " + "\t" + book.getName());
            System.out.println("Author : " + "\t" + book.getAuthor());
            System.out.println("Rank : " + "\t" + book.getRank());
            System.out.println("Status : " + "\t" + book.getStatus());
        }
    }
}

```

//Sort

```

public void sortByRank() {

    for (int j = 0; j < bookCount - 1; j++) {
        for (int k = 0; k < bookCount - 1; k++) {

            if (bookList[j].getRank() < bookList[k].getRank()) {
                Book temp = bookList[j];
                bookList[j] = bookList[k];

                bookList[k] = temp;
            }
        }
    }

    //printing Sorted array
    Book[] sorted = bookList;
    System.out.println("ID" + "\tName" + "\t\tAuthor(s)" + "\tRank" + "\tStatus");
    for (int l = 0; l <= bookCount - 1; l++) {

```

```

        sorted[l].setId(l + 1);
        System.out.println(sorted[l].getId() + "\t" + sorted[l].getName() + "\t\t" +
sorted[l].getAuthor() + "\t"
            + sorted[l].getRank() + "\t" + sorted[l].getStatus());
    }

}

}

```

```

public class Driver {
    public static void main(String[] args) {

        BookList My_Booklist = new BookList("My Booklist");

        My_Booklist.AddBook("Uncanny Valley", "Anna Wiener", 10);
        My_Booklist.AddBook("Weather", "Jenny Offil", 4);
        My_Booklist.AddBook("Long Bright River", "Liz Moore", 18);
        My_Booklist.AddBook("The Glass Hotel", "Emily St & John Mandel", 2);
        My_Booklist.AddBook("Afterlife", "Julia Alvarez", 14);

        My_Booklist.searchBook("Long Bright River").setRank(1);

        My_Booklist.searchBook("Afterlife").setStatus("READING");

        My_Booklist.DisplayList();

        System.out.println("#####");
        System.out.println("");
        My_Booklist.sortByRank();

        System.out.println("#####");
        System.out.println("");

        My_Booklist.DeleteBook(3);
        My_Booklist.DisplayList();

    }
}

```

Q2.

```
public class Genere {
    protected int ID;
    protected Artist[] artists = new Artist[10];
    protected int hourly_rate;
    protected int artist_count = 0;

    protected void addWork(int artist_id, int no_of_songs) {
        this.artists[this.artist_count] = new Artist(artist_id, no_of_songs);
        this.artist_count++;
    }

    protected Artist geArtist(int artist_id) {
        return this.artists[artist_id];
    }

    protected int getPayment(int artist_no) {

        for (int i = 0; i < this.artist_count; i++) {
            if (this.artists[i].getArtistId() == artist_no) {
                return this.artists[i].getNumber_of_songs() * hourly_rate;
            }
        }
        return 0;
    }
}
```

```
public class Artist {

    private int artistId;
    private int number_of_songs;

    public Artist(int id, int songs) {
        this.artistId = id;
        this.number_of_songs = songs;
    }

    public int getArtistId() {
        return artistId;
    }

    public int getNumber_of_songs() {
        return number_of_songs;
    }
}
```

```
}  
}
```

```
public class DeathMetal extends Genre {  
  
    public DeathMetal() {  
        this.ID = 5;  
        this.hourly_rate = 700;  
    }  
  
}
```

```
public class EDM extends Genre {  
  
    public EDM() {  
        this.ID = 2;  
        this.hourly_rate = 800;  
    }  
  
}
```

```
public class HipHop extends Genre {  
  
    public HipHop() {  
        this.ID = 3;  
        this.hourly_rate = 400;  
    }  
  
}
```

```
public class Jazz extends Genre {  
  
    public Jazz() {  
        this.ID = 4;  
        this.hourly_rate = 500;  
    }  
  
}
```

```
public class Rock extends Genere {
```

```
    public Rock() {
```

```
        this.ID = 1;
```

```
        this.hourly_rate = 600;
```

```
    }
```

```
}
```

```
public class Driver {
```

```
    public static void main(String[] args) throws Exception {
```

```
        Rock rock = new Rock();
```

```
        EDM edm = new EDM();
```

```
        HipHop hipHop = new HipHop();
```

```
        Jazz jazz = new Jazz();
```

```
        DeathMetal deathMeatal = new DeathMetal();
```

```
        Genere[] genres = { rock, edm, hipHop, jazz, deathMeatal };
```

```
        int[] artist_nos = { 1001, 2001, 3001, 4001, 5001, 6001 };
```

```
        rock.addWork(1001, 8);
```

```
        rock.addWork(2001, 4);
```

```
        rock.addWork(3001, 3);
```

```
        edm.addWork(2001, 7);
```

```
        edm.addWork(3001, 5);
```

```
        edm.addWork(1001, 6);
```

```
        hipHop.addWork(1001, 5);
```

```
        hipHop.addWork(2001, 6);
```

```
        hipHop.addWork(4001, 4);
```

```
        jazz.addWork(6001, 8);
```

```
        jazz.addWork(4001, 6);
```

```
        jazz.addWork(5001, 7);
```

```
        deathMeatal.addWork(5001, 5);
```

```
        deathMeatal.addWork(3001, 7);
```

```
        deathMeatal.addWork(6001, 4);
```

```
        for (int n : artist_nos) {
```

```
            int salary = 0;
```

```
            for (Genere g : genres) {
```

```
                salary = g.getPayment(n) + salary;
```

```
    }  
    System.out.println(n);  
    System.out.println("Total Salary:" + salary);  
}  
  
}  
}
```